Dragon Air: maths record

Dragon Air · Chapter 1 · 9–15 September · by kind of maths · printed 13 October 2026

Litres × price per litre

NCP-3-06 · NCP-3-05 · ALG-3-08 · FDP-3-06

Week 2 · Monday 9 September 2030 · Fuel invoice: Saab 340 on Dublin NCP-3-06 · NCP-3-05 · ALG-3-08 · FDP-3-06

Check before payment: 230 litres × £1.47 (Calculate)

Attempt 1

100 L costs = 1
200 L costs =
30 L costs =
230 L × £1.47 =

Answer: (not filled in) Query

100 L costs £147, not £1.

Attempt 2

100 L costs = 147.00
200 L costs = 294.00
30 L costs = 44.10
230 L × £1.47 = 338.10

Answer: £338.10 ✓

Method: scale from 100 L · 2 attempts

Week's total and the mean

NCP-3-05 · STAT-3-19 · STAT-3-23 · MON-3-01 · PFA-MM-02

Week 2 · Sunday 15 September 2030 · Week 2 accounts NCP-3-05 · STAT-3-19 · STAT-3-23 · MON-3-01 · PFA-MM-02

Show the finance team how to work out the week's total. Mon 9 £1,465, Tue 10 £1,465, Wed 11 £1,465, Thu 12 £522, Fri 13 £1,427, Sat 14 £1,091, Sun 15 £937 (Build)

Mon 9 £1,465 + Tue 10 £1,465 + Wed 11 £1,465 + Thu 12 £522 + Fri 13 £1,427 + Sat 14 £1,091 + Sun 15 £937 = £8,372

✓

Method: show what to add · 1 attempt

Week 2 · Sunday 15 September 2030 · Week 2 accounts: Planned against what happened NCP-3-05 · STAT-3-19 · STAT-3-23 · MON-3-01 · PFA-MM-02

Show how to find the difference between what we planned and what happened. Projected £9,493, Actual £8,372 (Build)

Projected £9,493 − Actual £8,372 = £1,121 less than planned

✓

Method: larger − smaller · 1 attempt

Week 2 · Sunday 15 September 2030 · Week 2 accounts: Average profit a day NCP-3-05 · STAT-3-19 · STAT-3-23 · MON-3-01 · PFA-MM-02

What was our average profit a day this week? Week's total £8,372 shared between the days (Build)

`Week's total £8,372 ÷ 7 = £1,196 a day`

✓

Method: show the divisor (total ÷ number) · 1 attempt

Fuel purchase

NCP-3-06 · NCP-3-02 · MON-3-01 · PFA-MM-02

Week 2 · Monday 9 September 2030 · Fuel order: the first order on the desk NCP-3-06 · NCP-3-02 · MON-3-01 · PFA-MM-02

Sign off the order: 3 blocks of 1,000 L at £1.47. 1,000 L × £1.47, then × 3 (Calculate)

Attempt 1

`1,000 L × £1.47 = 1`
`3 × £1,470.00 =`

Answer: (not filled in) Query

1,000 L × £1.47 is £1,470, not £1.

Attempt 2

`1,000 L × £1.47 = 1470.00`
`3 × £1,470.00 = 4410.00`

Answer: £4,410.00 ✓

Method: cost of 1,000 L × number of blocks · 2 attempts

Price ÷ rent

NCP-3-06 · MON-3-03 · PFA-MM-07

Week 2 · Monday 9 September 2030 · Rent or buy: Saab 340 NCP-3-06 · MON-3-03 · PFA-MM-07

How many days of rent equal the price? £22,000 ÷ £400 a day (Calculate)

Attempt 1

Answer: 50 days Query

Answer 50 days given; correct answer 55 days.

Attempt 2

Answer: 55 days ✓

Method: short division · 2 attempts

Decisions, week by week

Week 2 · 9–15 September · Decisions

Aircraft	Flights
Twin Otter G-DRAG	Paris 07:00 · Paris 18:00
Saab 340 G-DRAA	Dublin 08:00 · Dublin 16:00

Fares: Paris £90 · Dublin £80 · On board: high

Changed this week: Dublin: 1 → 2 flights a day; Fare: Dublin £90 → £80; New aircraft: Saab 340 G-DRAA

Events: Crew sickness: Cancel the flight

Aircraft: Saab 340 G-DRAA (rented at £400 a day)

Fuel bought: 3,000 L (£4,410.00) · fuel purchasing advantage −£178

Results: projected £9,493 · actual £8,372 · missed £1,421 · stars ★★★ (3)